TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Fibre Properties by AFIS and HVI

Two automated philosophies side by side: HVI tests fibre bundles at bale speed for classing and laydown control, while AFIS individualises thousands of single fibres to expose neps, short fibres and maturity that bundle tests average away.

TE-3104 · QUALITY EVALUATION OF FIBRE AND YARN SESSIONAL · EXPERIMENT 3 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to measure the fibre properties of a cotton sample on the High Volume Instrument (HVI) and on the Advanced Fibre Information System (AFIS), to tabulate and interpret every reported parameter, and to explain — from the measurement principle of each instrument — why the two systems agree on some properties and deliberately differ on others.

1.1 HVI: bundle testing at high volume

The HVI line was built for speed: a trained operator can class a bale sample in under a minute, which is why entire national crops are classed on it. Every HVI measurement is made on a **bundle** of fibres. A brushed beard of fibres is scanned by light to build a fibrogram from which length parameters are extracted; the same beard is broken between jaws at a fixed 1/8-inch gauge for strength; a separate plug is tested by airflow for micronaire; and a colour head measures reflected light for grade. The reported parameters are:

- **Micronaire (Mic)** — airflow fineness index, as in Experiment 1: a combined function of fineness and maturity.
- **Upper-half mean length (UHML)** — mean length of the longest half of the fibres by weight, reported in inches (and mm); the closest instrumental equivalent of staple length.
- **Uniformity index (UI)** — mean length ÷ UHML × 100. Above 85% is very uniform, 82–85% good, below 80% irregular with a heavy short-fibre tail.
- **Strength (Str)** — bundle breaking tenacity in g/tex at 1/8-inch gauge. Below 26 is weak, 27–29 medium, 30 and above strong.
- **Elongation (Elg)** — percentage extension of the bundle at break; higher elongation generally means better spinning performance and fewer ends down.

- **Colour (Rd and +b)** — Rd is reflectance (brightness/greyness, higher is brighter), +b is yellowness. Together they place the cotton in a colour grade quadrant of the classer's chart.

Because HVI works on bundles, it reports accurate averages very fast, but a bundle average cannot count individual defects: a few hundred neps or a tail of very short fibres barely moves the mean.

1.2 AFIS: fibre-by-fibre testing

AFIS takes the opposite approach. An opening roller and air separation **individualise** the sample so that fibres pass one at a time through electro-optical sensors; typically 3,000–5,000 fibres or particles are counted per replicate. Each event is classified by its signal shape, which lets AFIS *count* things HVI can only average:

- **Neps** — total nep count per gram, split into fibre neps and seed-coat neps, with mean nep size in micrometres — the core subject of Experiment 4.
- **Length by number, L(n), and by weight, L(w)** — because each fibre is counted, AFIS gives the true number distribution. L(n) is always shorter than L(w) because the numerous short fibres dominate the count but contribute little mass. Their spread gives the length CV, and the proportion below 12.7 mm gives short fibre content by number, SFC(n) — far more sensitive than the HVI estimate.
- **Trash and dust** — counted particle by particle: total trash count per gram, dust count (particles below 500 µm), and the visible foreign matter (VFM) percentage.
- **Fineness and maturity** — from the individual fibre signals AFIS reports fineness in mtex and a maturity ratio derived from fibre cross-section development, independent of the airflow principle.

> **THE ONE-SENTENCE DISTINCTION**
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> HVI asks: *what is this bale like on average, right now, fast enough to class a crop?* AFIS asks: *what is actually in this sample, fibre by fibre — including the defects that will become yarn faults?* A modern quality laboratory runs both: HVI on every bale, AFIS on laydowns and on each process stage to watch neps and short fibres rise or fall through the mill.

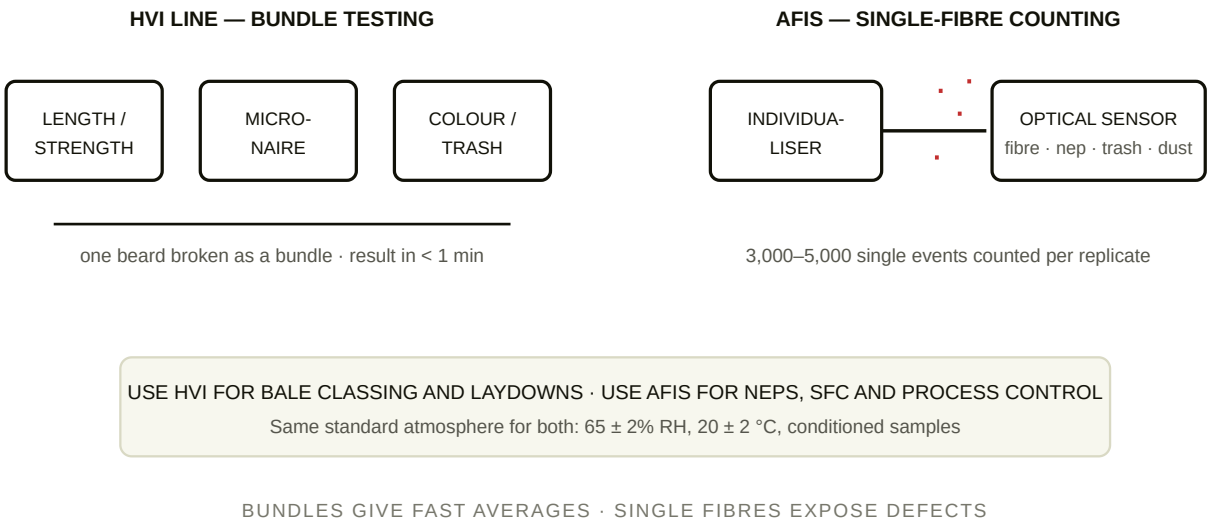


Figure 1 — Measurement architecture of the two systems: HVI modules test fibre bundles (left), while AFIS individualises the sample and classifies single fibres, neps, trash and dust at an optical sensor (right).

2. Apparatus and procedure

- **Apparatus:** HVI line (length/strength, micronaire and colour/trash modules) with calibration cottons and colour tiles; AFIS with its fibre individualiser module and nep/trash and fineness/maturity sensor options; balances; conditioned laboratory samples (minimum 24 h at 65 ± 2% RH, 20 ± 2 °C); sample preparation combs and brushes.
- **Calibration first:** verify HVI with the two standard calibration cottons (short/weak and long/strong) and the colour tile set; verify AFIS with its reference sample. No sample result is recorded until calibration values fall inside their tolerance bands — both instruments drift, and an uncalibrated average is worse than no average.
- **HVI run:** prepare a representative handful (about 10 g for length/strength, the specified plug for micronaire). Form the beard on the sampling comb, load, and run the sequence: UHML, UI, strength and elongation from the broken beard; micronaire from the airflow plug; Rd and +b from the colour window, which also images the surface for a trash estimate. Test at least two replicates per bale sample.
- **AFIS run:** weigh the specified subsample (0.5 g per replicate is typical), feed it to the individualiser, and let the system count its full replicate of single events. Run at least three replicates and average; watch the replicate scatter — a wide spread between replicates signals poor mixing of the laboratory sample, not instrument noise.
- **Reporting:** tabulate HVI and AFIS results side by side, convert UHML to millimetres (× 25.4), and interpret each parameter against its commercial band before writing the laydown recommendation.

3. Worked calculation

HVI results (model data). The bale sample returns:

PARAMETER	RESULT	INTERPRETATION
Micronaire	4.3	Medium-fine, in the premium 3.7–4.8 band; no discount.
UHML length	1.14 in = 28.96 mm	Medium-long staple; supports counts to about 40 Ne in warp.
Uniformity index	82.5%	Good uniformity (82–85 band); a modest short-fibre tail.
Strength	29.8 g/tex	Upper end of medium, bordering on strong (30); good yarn strength expected.
Elongation	6.8%	Healthy; fibres yield rather than snap in drafting.
Colour Rd	76.5	Bright — high reflectance, a white grade.
Colour +b	8.2	Moderate yellowness; together with Rd 76.5 this places the cotton in a good middling white grade region.

UHML conversion: 1.14 in × 25.4 = 28.96 mm

AFIS results (same laydown). A consistent fibre-by-fibre picture of the same cotton:

AFIS PARAMETER	RESULT	INTERPRETATION
Total neps	248 /g (fibre neps 221, seed-coat neps 27)	Moderate raw-cotton nep load; carding must remove the large majority (see Experiment 4).
Mean nep size	720 µm (0.72 mm)	Typical; large enough to be card-removable, small neps would pass through.
Length by weight L(w)	26.8 mm	Consistent with the HVI length once the different principles are allowed for.
Length by number L(n)	21.4 mm	Shorter than L(w), as it must be: short fibres dominate the count.
Short fibre content SFC(n)	11.8%	Moderate; watch waste and yarn evenness at fine counts.

Trash count / dust count	68 /g / 412 /g	Matches the 3.9% Shirley trash of Experiment 1 in character: thorough blowroom cleaning required.
Fineness / maturity ratio	168 mtex / 0.87	Confirms micronaire 4.3 as genuine fineness — maturity is sound (compare 165 mtex and 0.85 by the manual methods of Experiment 1).

The two instruments corroborate each other where they overlap — length, fineness and maturity all agree within the expected method differences — and complement each other where they do not: only AFIS counts the 248 neps/g and the 11.8% short fibres by number that will govern card settings and yarn faults.

> **LAYDOWN DECISION**
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> This cotton — 28.96 mm, UI 82.5%, 29.8 g/tex, micronaire 4.3, maturity 0.87 — is a sound medium-long mixing for 30 Ne warp/weft of the kind characterised in Experiments 2 and 5. Its nep count and trash demand careful blowroom and card settings, not rejection.

4. Precautions

- Calibrate before testing and re-check at the interval the laboratory protocol sets; HVI strength and micronaire are comparative measurements that are only as true as their calibration cottons.
- Condition every sample fully. Moisture changes fibre mass (micronaire), bundle strength and colour readings simultaneously — an unconditioned sample disagrees with itself across modules.
- Mix the laboratory sample thoroughly before taking either instrument's subsample. AFIS consumes only about 0.5 g per replicate; an unmixed pinch can be all leaf or all clean lint.
- Prepare the HVI beard evenly — thin or lumpy beards break low and read short. Keep hands off the beard surface: skin oil changes both strength and colour readings.
- Never compare AFIS L(n) directly with HVI UHML as if they were the same statistic; one is a number mean of all fibres, the other a weight-biased mean of the longest half. Compare like with like, or state the basis explicitly.
- Clean the AFIS individualiser and sensor windows between samples; carry-over neps and dust from a dirty sample inflate the next sample's counts.

Practice (viva preparation)

1. State the fundamental measurement difference between HVI and AFIS, and name one property each instrument measures that the other cannot.

2. Interpret this HVI line in full: Mic 4.3, UHML 1.14 in, UI 82.5%, Str 29.8 g/tex, Elg 6.8%, Rd 76.5, +b 8.2.
3. Why is AFIS length by number always shorter than length by weight for the same cotton?
4. A cotton reads micronaire 3.2 on HVI. What does AFIS add to the decision on this cotton?
5. Why do mills class every bale on HVI but run AFIS only on selected samples and process stages?
6. What do Rd and +b each measure, and why are both needed for a colour grade?

Answer key

1. HVI measures properties of fibre bundles — a beard is scanned and broken as one unit, a plug is tested by airflow. AFIS individualises the sample and classifies thousands of single fibres and particles optically. Only AFIS can count neps per gram (and split fibre neps from seed-coat neps) and give true short fibre content by number; only HVI gives the standardised colour grade (Rd, +b) and bale-classing bundle strength at crop speed.
2. Micronaire 4.3: medium-fine, premium band. UHML 1.14 in = 28.96 mm: medium-long staple. UI 82.5%: good uniformity. Strength 29.8 g/tex: top of medium, nearly strong. Elongation 6.8%: healthy spinning behaviour. Rd 76.5: bright; +b 8.2: moderately yellow — together a good white grade. Overall: a sound mixing for medium counts such as 30 Ne.
3. Short fibres are numerous but light. A number average gives every fibre one vote, so the many short fibres pull L(n) down; a weight average gives votes in proportion to mass, which the long fibres dominate. The gap between L(w) and L(n) is itself a measure of how heavy the short-fibre tail is.
4. A micronaire of 3.2 may be genuinely fine fibre or immature fibre — HVI airflow cannot separate the two. AFIS reports fineness in mtex and a maturity ratio from single fibres, so it shows directly whether the lot is fine-and-mature (valuable) or immature (nep-prone, dye-risky, likely to be discounted or rejected).
5. Cost and speed. An HVI result takes under a minute, so 100% bale testing is feasible and necessary for laydowns and payment. An AFIS replicate takes minutes and a fraction of a gram, so it is used where its unique information pays: qualifying mixings, setting and auditing cards, tracing nep generation through the process line.
6. Rd is the percentage reflectance — how bright or grey the cotton appears. +b is its position on the yellow axis — how creamy or yellow it is. A cotton can be bright but yellow, or dull but white; grade charts are two-dimensional (the Nickerson–Hunter diagram), so both coordinates are needed to place a sample in a grade box.
